

VOIP

...

Veille

Le VoIP (Voice over Internet Protocol) est une technologie qui permet de transmettre la voix (et autre communications) sur des réseaux IP (Internet) au lieu des réseaux téléphonique traditionnels.

Avantages:

- Coût réduit (pas de lignes téléphonique classique)
- Flexibilité (appels depuis n'importe où)
- Fonctionnalités avancées (transfert, conférence, messagerie vocale...)

- SIP (Session Initiation Protocol)

Rôle : Protocole de signalisation qui gère :

- L'établissement de la communication (quand tu composes un numéro)
- La modification (transfert d'appel)
- La terminaison (raccrocher)

Par défaut le port est à 5060

- RTP (Real-time Transport Protocol)

Rôle: Protocole de transport qui transporte le flux audio/vidéo en temps réel

- Minimise la latence
- Supporte la perte de paquets (certains peuvent être perdus sans couper la conversation)

Concepts de sécurité

Le mot de passe qui empêche les appels non autorisés.

Le chiffrement via SRTP = RTP chiffré (audio sécurisé), TLS = SIP chiffré (signalisation sécurisée)

Architecture du VoIP:

Composant	Rôle	Exemple
IPBX	Serveur téléphonique IP	Asterisk, FreePBX
User Agent (UA)	Client VoIP (tel app)	Linphone, Zoiper
Gateway	Passerelle entre VoIP	Permet d'appeler
RTC classique	des flux	Registrar
Serveur qui enregistre les utilisateurs		Fonction d'Asterisk
Proxy SIP	Routeur d'appels SIP	Peut être Asterisk

Installation et configuration

```
root@debian:~# cd /usr/src
root@debian:/usr/src# wget https://downloads.asterisk.org/pub/telephony/asterisk/asterisk-22-current.tar.gz
--2026-02-18 12:03:48-- https://downloads.asterisk.org/pub/telephony/asterisk/asterisk-22-current.tar.gz
Résolution de downloads.asterisk.org (downloads.asterisk.org)... 165.22.184.19, 2604:a880:400:d0::14:9001
Connexion à downloads.asterisk.org (downloads.asterisk.org)|165.22.184.19|:443... connecté.
requête HTTP transmise, en attente de la réponse... 200 OK
Taille : 26523526 (25M) [application/octet-stream]
Sauvegarde en : « asterisk-22-current.tar.gz »

asterisk-22-current.ta 100%[=====] 25,29M 2,57MB/s ds 7,0s

2026-02-18 12:03:56 (3,64 MB/s) - « asterisk-22-current.tar.gz » sauvegardé [26523526/26523526]
```

```
root@debian:/usr/src# cd asterisk-22.*/
root@debian:/usr/src/asterisk-22.8.2# contrib/scripts/install_prereq install
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Les paquets supplémentaires suivants seront installés :
  aptitude-common libboost-iostreams1.83.0 libcwwidget4 libsigc++-2.0-0v5 libxapian30
Paquets suggérés :
  apt-xapian-index aptitude-doc-en | aptitude-doc debtags libcwwidget-dev xapian-tools
Les NOUVEAUX paquets suivants seront installés :
  aptitude aptitude-common libboost-iostreams1.83.0 libcwwidget4 libsigc++-2.0-0v5
```

```
root@debian:/usr/src/asterisk-22.8.2# ./configure
checking build system type... x86_64-pc-linux-gnu
checking host system type... x86_64-pc-linux-gnu
checking for gcc... gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
```

```

root@debian:/usr/src/asterisk-22.8.2# sudo make menuselect
-bash: sudo : commande introuvable
root@debian:/usr/src/asterisk-22.8.2# make menuselect
make[1] : on entre dans le répertoire « /usr/src/asterisk-22.8.2 »
[jansson] Cleaning
[pjproject] Cleaning
[libjwt] Cleaning
rm -f defaults.h
rm -f include/asterisk/build.h

```



```

Add-ons (See README-addons.txt)
Applications
Bridging Modules
Call Detail Recording
Channel Event Logging
Channel Drivers
Codec Translators
Format Interpreters
Dialplan Functions
PBX Modules
Resource Modules
Test Modules
Alternate Channel Storage Backend
Compiler Flags
Utilities
AGI Samples
Core Sound Packages
Music On Hold File Packages
Extras Sound Packages

```

```

Channel Drivers
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Test Modules
Alternate Channel Storage Backend
Compiler Flags
Utilities
AGI Samples
Core Sound Packages

```

```

root@debian:/usr/src/asterisk-22.8.2# make -j$(nproc)
CC="cc" CXX="g++" LD="" AR="" RANLIB="" CFLAGS="" LDFLAGS="" make -C menuselect CONFIGU
RE_SILENT="--silent" makeopts
make[1] : on entre dans le répertoire « /usr/src/asterisk-22.8.2/menuselect »
make[1]: « makeopts » est à jour.
make[1] : on quitte le répertoire « /usr/src/asterisk-22.8.2/menuselect »
[CC] astcanary.c -> astcanary.o
[CC] astdb2sqlite3.c -> astdb2sqlite3.o
[CC] hash/hash.c -> hash/hash.o
[CC] astdb2bdb.c -> astdb2bdb.o
[LD] astcanary.o -> astcanary

```

```

[LD] res_websocket_client.o -> res_websocket_client.so
[LD] res_xmpp.o -> res_xmpp.so
[CC] format_mp3.c -> format_mp3.o
[CC] mp3/common.c -> mp3/common.o
[CC] mp3/dct64_i386.c -> mp3/dct64_i386.o
[CC] mp3/decode_ntom.c -> mp3/decode_ntom.o
[CC] mp3/layer3.c -> mp3/layer3.o
[CC] mp3/tabininit.c -> mp3/tabininit.o
[CC] mp3/interface.c -> mp3/interface.o
[LD] format_mp3.o mp3/common.o mp3/dct64_i386.o mp3/decode_ntom.o mp3/layer3.o mp3/t
abininit.o mp3/interface.o -> format_mp3.so
Building Documentation For: channels pbx apps codecs formats cdr cel bridges funcs test
s main res addons
+----- Asterisk Build Complete -----+
+ Asterisk has successfully been built, and +
+ can be installed by running: +
+ +
+ make install +
+-----+

```

```

root@debian:/usr/src/asterisk-22.8.2# make install
CC="cc" CXX="g++" LD="" AR="" RANLIB="" CFLAGS="" LDFLAGS="" make -C menuselect CONFIGU
RE_SILENT="--silent" makeopts
make[1] : on entre dans le répertoire « /usr/src/asterisk-22.8.2/menuselect »
make[1]: « makeopts » est à jour.
make[1] : on quitte le répertoire « /usr/src/asterisk-22.8.2/menuselect »
Installing modules from channels...
Installing modules from pbx...
Installing modules from apps...
Installing modules from codecs...
codec_opus: Remote package version 22.0_1.3.0 (263680)
codec_opus: Downloading https://downloads.digium.com/pub/telephony/codec_opus/asterisk-
22.0/x86-64/codec_opus-22.0_1.3.0-x86_64.tar.gz to /tmp/tmp.rPhR0zCmfJ/codec_opus-22.0_
1.3.0-x86_64.tar.gz

```

```

done
+---- Asterisk Installation Complete -----+
+
+ YOU MUST READ THE SECURITY DOCUMENT +
+
+ Asterisk has successfully been installed. +
+ If you would like to install the sample +
+ configuration files (overwriting any +
+ existing config files), run: +
+
+ For generic reference documentation: +
+ make samples +
+
+ For a sample basic PBX: +
+ make basic-pbx +
+
+
+----- or -----+
+
+ You can go ahead and install the asterisk +
+ program documentation now or later run: +
+
+ make progdocs +
+
+ **Note** This requires that you have +
+ doxygen installed on your local system +
+-----+

```

```
root@debian:/usr/src/asterisk-22.8.2# make config
root@debian:/usr/src/asterisk-22.8.2# ldconfig
root@debian:/usr/src/asterisk-22.8.2#
```

```
root@debian:/usr/src/asterisk-22.8.2# systemctl start asterisk
```

```
root@debian:/usr/src/asterisk-22.8.2# systemctl status asterisk
● asterisk.service - LSB: Asterisk PBX
   Loaded: loaded (/etc/init.d/asterisk; generated)
   Active: active (running) since Thu 2026-02-19 10:30:25 CET; 21s ago
 Invocation: cbfc7124374547b1aa6b7159daf8ff82
    Docs: man:systemd-sysv-generator(8)
   Process: 28324 ExecStart=/etc/init.d/asterisk start (code=exited, status=0/SUCCESS)
    Tasks: 73 (limit: 2272)
   Memory: 65.1M (peak: 67.2M)
      CPU: 3.633s
   CGroup: /system.slice/asterisk.service
           └─28336 /usr/sbin/asterisk
```

[illegible]

```
root@debian:/usr/src/asterisk-22.8.2# asterisk -rvvvvv
Asterisk 22.8.2, Copyright (C) 1999 - 2025, Sangoma Technologies Corporation and others
.
Created by Mark Spencer <markster@digium.com>
Asterisk comes with ABSOLUTELY NO WARRANTY; type 'core show warranty' for details.
This is free software, with components licensed under the GNU General Public
License version 2 and other licenses; you are welcome to redistribute it under
certain conditions. Type 'core show license' for details.
=====
Connected to Asterisk 22.8.2 currently running on debian (pid = 28336)
debian*CLI> |
```

```
root@debian:/usr/src/asterisk-22.8.2# contrib/scripts/get_mp3_source.sh
A   addons/mp3
A   addons/mp3/layer3.c
A   addons/mp3/interface.c
A   addons/mp3/MPGLIB_TODO
A   addons/mp3/mpg123.h
A   addons/mp3/mpglib.h
A   addons/mp3/decode_ntom.c
A   addons/mp3/MPGLIB_README
A   addons/mp3/common.c
A   addons/mp3/huffman.h
A   addons/mp3/tabininit.c
A   addons/mp3/Makefile
A   addons/mp3/README
A   addons/mp3/decode_i386.c
A   addons/mp3/dct64_i386.c
Exporté à la révision 204.
```

```

root@debian:/# asterisk -rx "pjsip reload"
Module 'res_pjsip.so' reloaded successfully.
Module 'res_pjsip_authenticator_digest.so' reloaded successfully.
Module 'res_pjsip_endpoint_identifier_ip.so' reloaded successfully.
Module 'res_pjsip_mwi.so' reloaded successfully.
Module 'res_pjsip_notify.so' reloaded successfully.
Module 'res_pjsip_outbound_publish.so' reloaded successfully.
Module 'res_pjsip_publish_asterisk.so' reloaded successfully.
Module 'res_pjsip_outbound_registration.so' reloaded successfully.

```

```

; extensions.conf - the Asterisk dial plan
[default]
exten => 6001,1,Dial(PJSIP/6001,20)
exten => 6001,2,Hangup()

exten => 6002,1,Dial(PJSIP/6002,20)
exten => 6002,2,Hangup()

```

```

root@debian:/# asterisk -rx "dialplan reload"
Dialplan reloaded.

```

```

root@debian:~# sudo apt -y install build-essential git wget curl subversion automake autoconf libtool pkg-config unzip cmake
-bash: sudo : commande introuvable
root@debian:~# sudo apt -y install build-essential git wget curl subversion automake autoconf libtool pkg-config unzip cmake
-bash: sudo : commande introuvable
root@debian:~# sudo apt -y install build-essential git wget curl subversion automake autoconf libtool \
pkg-config unzip cmake
-bash: sudo : commande introuvable
root@debian:~# apt -y install build-essential git wget curl subversion automake autoconf libtool \
pkg-config unzip cmake
wget est déjà la version la plus récente (1.25.0-2).
Installation de :
  autoconf  build-essential  curl  libtool  subversion
  automake  cmake             git   pkg-config  unzip

Installation de dépendances :
autotools-dev      libbinutils      libnghttp3-9
binutils           libc-dev-bin     libngtcp2-16
binutils-common    libc6-dev        libngtcp2-crypto-gnutls8
binutils-x86-64-linux-gnu libcc1-0         libpkgconf3
cmake-data         libcrypt-dev     libquadmath0
cpp               libctf-nobfd0    libzstd1

```

```

root@debian:~# apt -y install libedit-dev uuid-dev libxml2-dev libssl-dev libncurses5-dev li
bsqlite3-dev \
libjansson-dev libxslt1-dev libsqlite3-dev libiksemel-dev libspeexdsp-dev \
liblua5.2-dev libcurl4-openssl-dev libpopt-dev libpq-dev unixodbc-dev
Note : sélection de « libncurses-dev » au lieu de « libncurses5-dev »
Installation de :
  libcurl4-openssl-dev  libjansson-dev  libpopt-dev  libsqlite3-dev  libxslt1-dev
  libedit-dev          liblua5.2-dev  libpq-dev   libssl-dev      unixodbc-dev
  libiksemel-dev       libncurses-dev  libspeexdsp-dev  libxml2-dev    uuid-dev

Installation de dépendances :
  comerr-dev          libgssrpc4t64      libnghttp2-dev  libssh2-1-dev
  krb5-multidev       libidn2-dev        libnghttp3-dev  libtasn1-6-dev
  libbrotli-dev       libiksemel3        libodbc2        libtasn1-doc
  libbsd-dev          libkadm5clnt-mit12 libodbc2        libtool-bin
  libevent-2.1-7t64   libkadm5srv-mit12 libodbcinst2    libunbound8
  libgmp-dev          libkdb5-10t64     libp11-kit-dev  libzstd-dev
  libgmpxx4ldbl       libkrb5-dev        libpq5          nettle-dev
  libgnutls-dane0t64  libldap-dev        libpsl-dev      unixodbc-common
  libgnutls-openssl27t64 liblua5.2-0        libreadline-dev zlib1g-dev
  libgnutls28-dev     libmd-dev          librtmp-dev
  libgpm2             libncurses6        libspeexdsp1

Paquets suggérés :
  doc-base      libgmp10-doc  gpm          tdsodbc      speex-doc
  krb5-doc       libmpfr-dev   krb5-user    p11-kit-doc  sqlite3-doc
  libcurl4-doc  dns-root-data ncurses-doc  postgresql-doc-17 libssl-doc

```

```

root@debian:~# apt -y install sngrep tcpdump http ufw fail2ban
Installation de :
  fail2ban  http  sngrep  tcpdump  ufw

Installation de dépendances :
  iptables          python3-infect      python3-pyinotify
  libip4tc2         python3-jaraco.context  python3-setuptools
  libip6tc2         python3-jaraco.function  python3-systemd
  libnetfilter-conntrack3 python3-jaraco.text      python3-typeguard
  libnfnetlink0     python3-more-itertools  python3-typing-extensions
  libpcap0.8t64     python3-pkg-resources   python3-zipp
  python3-autocommand python3-pyasyncore       whois

```

```
root@debian:/usr/src# cd asterisk-22.*/
root@debian:/usr/src/asterisk-22.8.2# contrib/scripts/install_prereq install
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Les paquets supplémentaires suivants seront installés :
  aptitude-common libboost-iostreams1.83.0 libcwwidget4 libsigc++-2.0-0v5 libxapian30
Paquets suggérés :
  apt-xapian-index aptitude-doc-en | aptitude-doc debtags libcwwidget-dev xapian-tools
Les NOUVEAUX paquets suivants seront installés :
  aptitude aptitude-common libboost-iostreams1.83.0 libcwwidget4 libsigc++-2.0-0v5
```

```
Aucun fichier schéma trouvé : aucune action effectuée
Paramétrage de libglib2.0-dev:amd64 (2.84.4-3~deb12u1) :
Paramétrage de libgmime-3.0-dev:amd64 (3.2.15+dfsg-1) :
```

```
#####
## install completed successfully
#####
root@debian:/usr/src/asterisk-22.8.2# |
```

```

root@debian:~# apt -y install libedit-dev uuid-dev libxml2-dev libssl-dev libncurses5-dev lib
bsqlite3-dev \
libjansson-dev libxslt1-dev libsqlite3-dev libiksemel-dev libspeexdsp-dev \
liblua5.2-dev libcurl4-openssl-dev libpopt-dev libpq-dev unixodbc-dev
Note : sélection de « libncurses-dev » au lieu de « libncurses5-dev »
Installation de :
  libcurl4-openssl-dev  libjansson-dev  libpopt-dev      libsqlite3-dev  libxslt1-dev
  libedit-dev          liblua5.2-dev  libpq-dev       libssl-dev      unixodbc-dev
  libiksemel-dev       libncurses-dev  libspeexdsp-dev  libxml2-dev     uuid-dev

Installation de dépendances :
  comerr-dev      libgssrpc4t64      libnghttp2-dev  libssh2-1-dev
  krb5-multidev   libidn2-dev        libnghttp3-dev  libtasn1-6-dev
  libbrotli-dev   libiksemel3        libodbc2        libtasn1-doc
  libbsd-dev      libkadm5clnt-mit12  libodbcrc2      libtool-bin
  libevent-2.1-7t64  libkadm5srv-mit12  libodbcinst2    libunbound8
  libgmp-dev       libkdb5-10t64      libp11-kit-dev  libzstd-dev
  libgmpxx4ldbl    libkrb5-dev        libpq5          nettle-dev
  libgnutls-dane0t64  libldap-dev        libpsl-dev      unixodbc-common
  libgnutls-openssl27t64  liblua5.2-0      libreadline-dev  zlib1g-dev
  libgnutls28-dev   libmd-dev          librtmp-dev
  libgpm2           libncurses6        libspeexdsp1

Paquets suggérés :
  doc-base      libgmp10-doc  gpm          tdsodbc      speex-doc
  krb5-doc      libmpfr-dev   krb5-user    p11-kit-doc  sqlite3-doc
  libcurl4-doc  dns-root-data  ncurses-doc  postgresql-doc-17  libssl-doc

```

```

root@debian:~# apt -y install sngrep tcpdump htop ufw fail2ban
Installation de :
  fail2ban  htop  sngrep  tcpdump  ufw

Installation de dépendances :
  iptables      python3-infect      python3-pyinotify
  libip4tc2     python3-jaraco.context  python3-setuptools
  libip6tc2     python3-jaraco.functools  python3-systemd
  libnetfilter-contrack3  python3-jaraco.text      python3-typeguard
  libnfnetlink0  python3-more-itertools    python3-typing-extensions
  libpcap0.8t64  python3-pkg-resources     python3-zipp
  python3-autocommand  python3-pyasyncore        whois

```

Utilisateurs

```
/etc/asterisk/pjsip.conf *
```

```
Samples and Quick Reference
```

```
[transport-udp]  
type=transport  
protocol=udp  
bind=0.0.0.0:5060
```

```
[6001]  
type=endpoint  
context=default  
disallow=all  
allow=ulaw  
allow=alaw  
auth=auth6001  
aors=6001
```

```
[auth6001]  
type=auth  
auth_type=userpass  
username=6001  
password=x1
```

```
[6001]  
type=aor  
max_contacts=1
```

```
root@debian:~# cat /etc/asterisk/extensions.conf | grep "\[default\]"
[default]
```

```
/etc/asterisk/extensions.conf *
sk dial plan
```

```
; extensions.conf - the Asterisk dial plan
[default]
exten => 6001,1,Dial(PJSIP/6001,20)
exten => 6001,2,Hangup()

exten => 6002,1,Dial(PJSIP/6002,20)
exten => 6002,2,Hangup()
```